

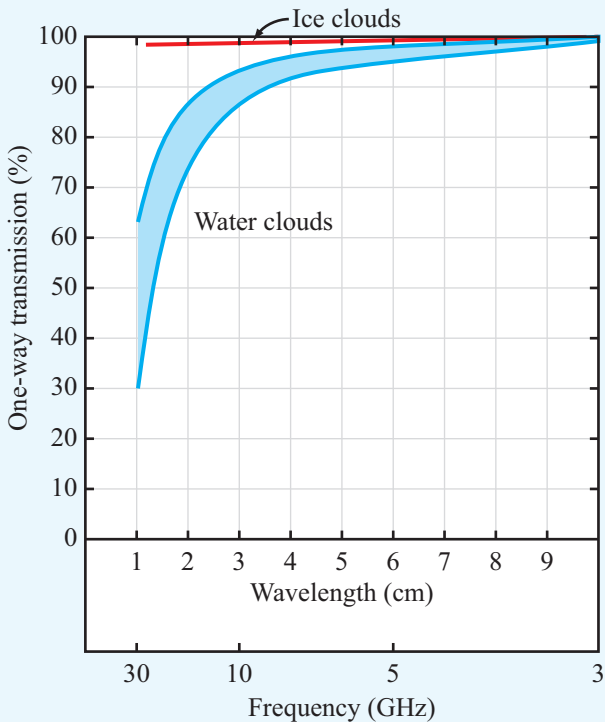


Figure 1-1: Attenuating effects of clouds on radio transmission from space to ground.

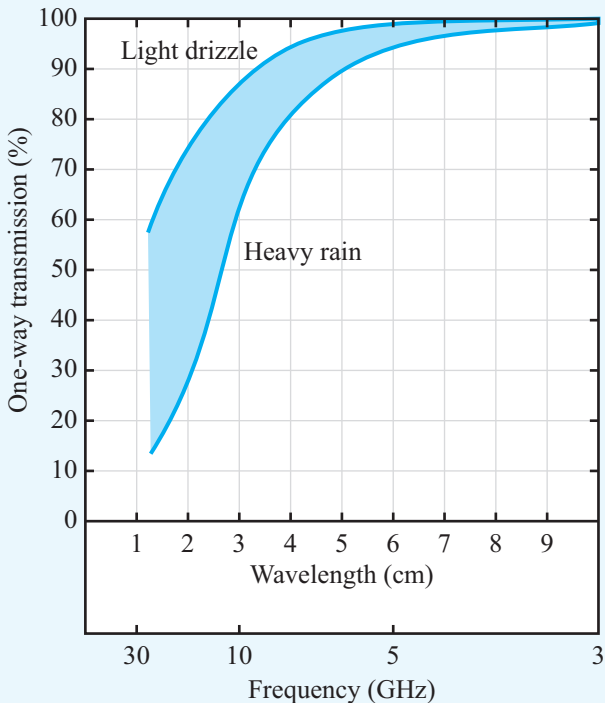
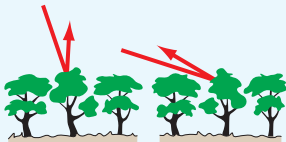


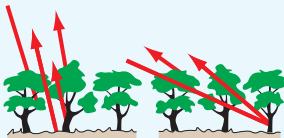
Figure 1-2: Attenuating effects of rain on radio transmission from space to ground.

1 cm wavelength



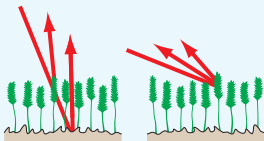
Radar signal
from tree crown

1 m wavelength



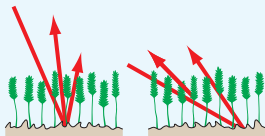
Signal from
crown, trunks,
ground

Signal from
crown, trunks



Signal from
wheat and
soil

Signal from
wheat



Signal from
soil and
subsoil

Signal from
wheat and
soil

Figure 1-3: Illustration of radar signal penetration through and scattering by vegetation. There is more penetration if vegetation is dry than if wet.

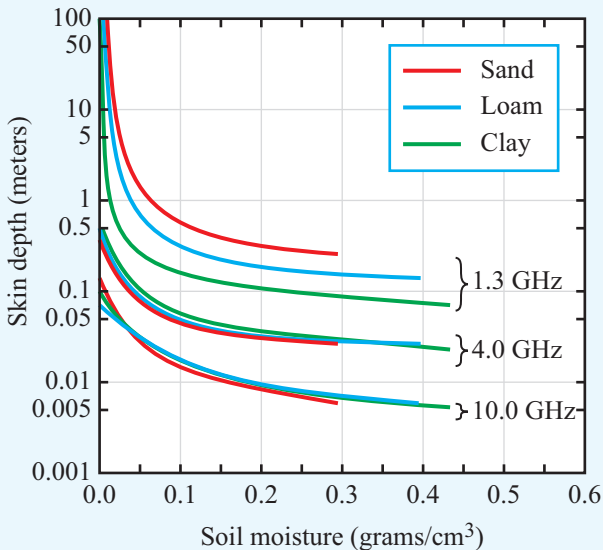


Figure 1-4: Skin depth as a function of soil volumetric water content, frequency, and soil type [from Ulaby et al., 1974].

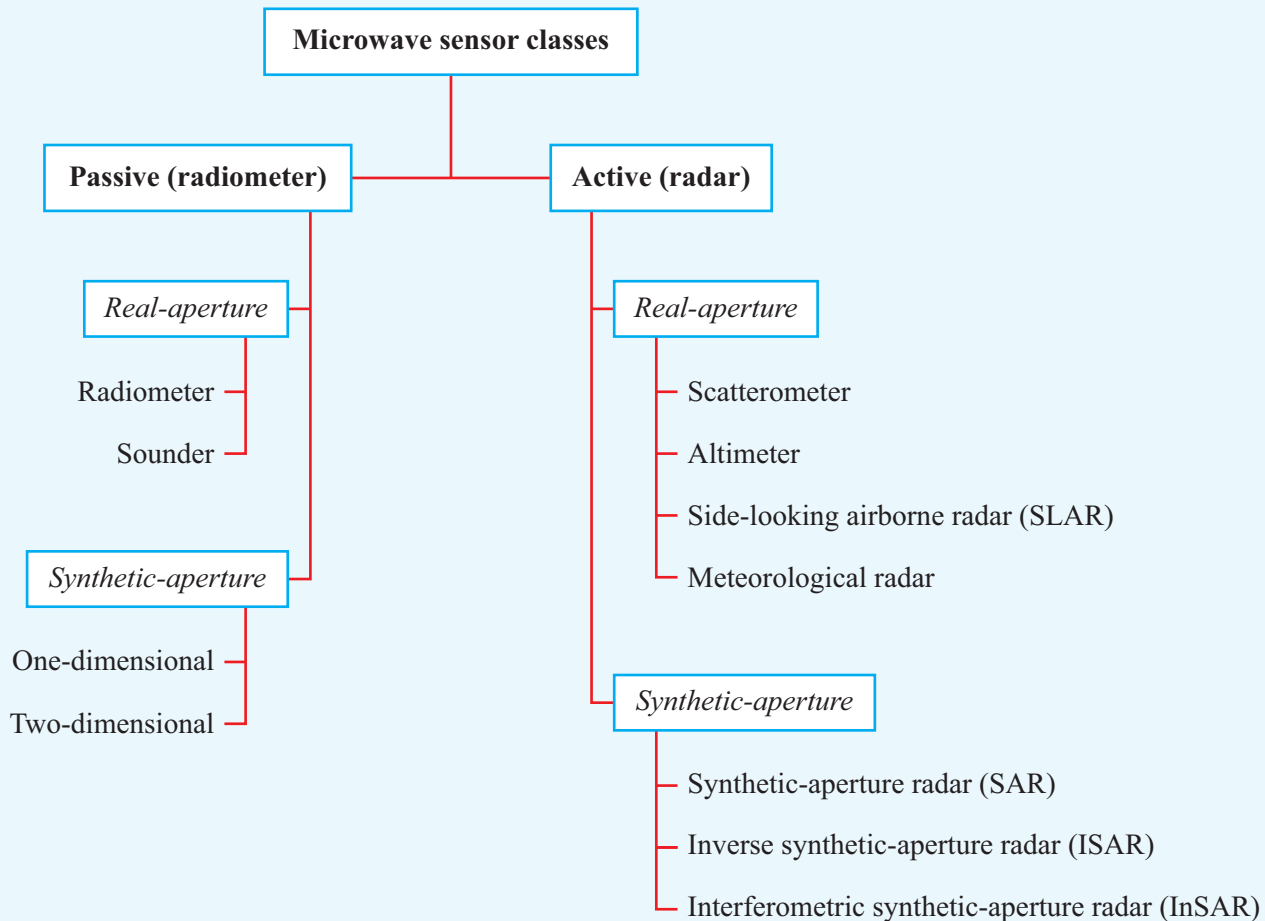


Figure 1-5: Major classes of microwave remote sensors.

Table 1-1: Overview of spaceborne SAR sensors and their main characteristics [Moreira et al., 2013].

Sensor	Operation	Frequency Band (Polarization)	Comments	Institution, Country
Seasat	1978	L (hh)	First civilian SAR satellite, operation for only ca. three months	NASA/JPL, USA
SIRA-A, SIR-B	1981, 1984	L (hh)	Shuttle radar missions	NASA/JPL, USA
ERS-1/2	1991–2000/ 1995–2011	C (vv)	European Remote Sensing Satellites (first European SAR satellites)	ESA, Europe
J-ERS-1	1992–1998	L (hh)	Japanese Earth Resource Satellite (first Japanese SAR satellite)	JAXA, Japan
SIR-C/X-SAR	April and October 1994	L & C (quad) X (vv)	Shuttle imaging radar mission, first demonstration of spaceborne multifrequency SAR	NASA/JPL, USA DLR, Germany ASI, Italy
Radarsat-1	1995–today	C (hh)	First Canadian SAR satellite, swath width of up to 500 km with Scan SAR imaging mode	CSA, Canada
SRTM	Feb. 2000	C (hh+vv) and X (vv)	Shuttle Radar Topography Mission, first spaceborne interferometric SAR	NASA/JPL, USA DLR, Germany ASI, Italy
ENVISAT/ASAR	2002–2012	C (dual)	First SAR satellite with Transmit/Receive module technology, swath width up to 400 km	ESA, Europe
ALOS/PALSAR	2006–2011	L (quad)	Advanced Land Observing Satellite (Daichi), swath width up to 360 km	JAXA, Japan
TerraSAR-X/ TanDEM-X	2007–today 2010–today	X (quad)	First bi-static radar in space, resolution up to 1 m, global topography available by end of 2014	DLR/Astrium, Germany
Radarsat-2	2007–today	C (quad)	Resolution up to: 1 m $\times$ 3 m (azimuth $\times$ range), swath width up to 500 km	CSA, Canada
COSMO-SkyMed-1/4	2007 ... 2010–today	X (dual)	Constellation of four satellites, up to 1 m resolution	ASI/MID, Italy
RISAT-1/2	2008–today	C (quad)	Follow-on satellite (RISAT-1a) to be launched in 2016, RISAT-3 (L-band) in development	ISRO, India
HJ-1C	2012–today	S (vv)	Constellation of four satellites, first satellite launched in 2012	CRESDA/CAST/ NRSCC, China
Kompsat-5	Launch scheduled in 2013	X (dual)	Korea Multi-Purpose Satellite 5, resolution up to 1 m	KARI, Korea
PAZ	Launch scheduled in 2013	X (quad)	Constellation with TerraSAR-X and TanDEM-X planned	CDTI, Spain
ALOS-2	Launch scheduled in 2013	L (quad)	Resolution up to: 1 m $\times$ 3 m (azimuth $\times$ range), swath width up to 490 km	JAXA, Japan
Sentinel-1a/1b	Launch scheduled in 2013/2015	C (dual)	Constellation of two satellites, swath width up to 400 km	ESA, Europe
Radarsat Constellation-1/2/3	Launch scheduled in 2017	C (dual)	Constellation of three satellites, swath width up to 500 km	CSA, Canada
SAOCOM-1/2	Launch scheduled in 2014/2015	L (quad)	Constellation of two satellites, fully polarimetric	CONAE, Argentina

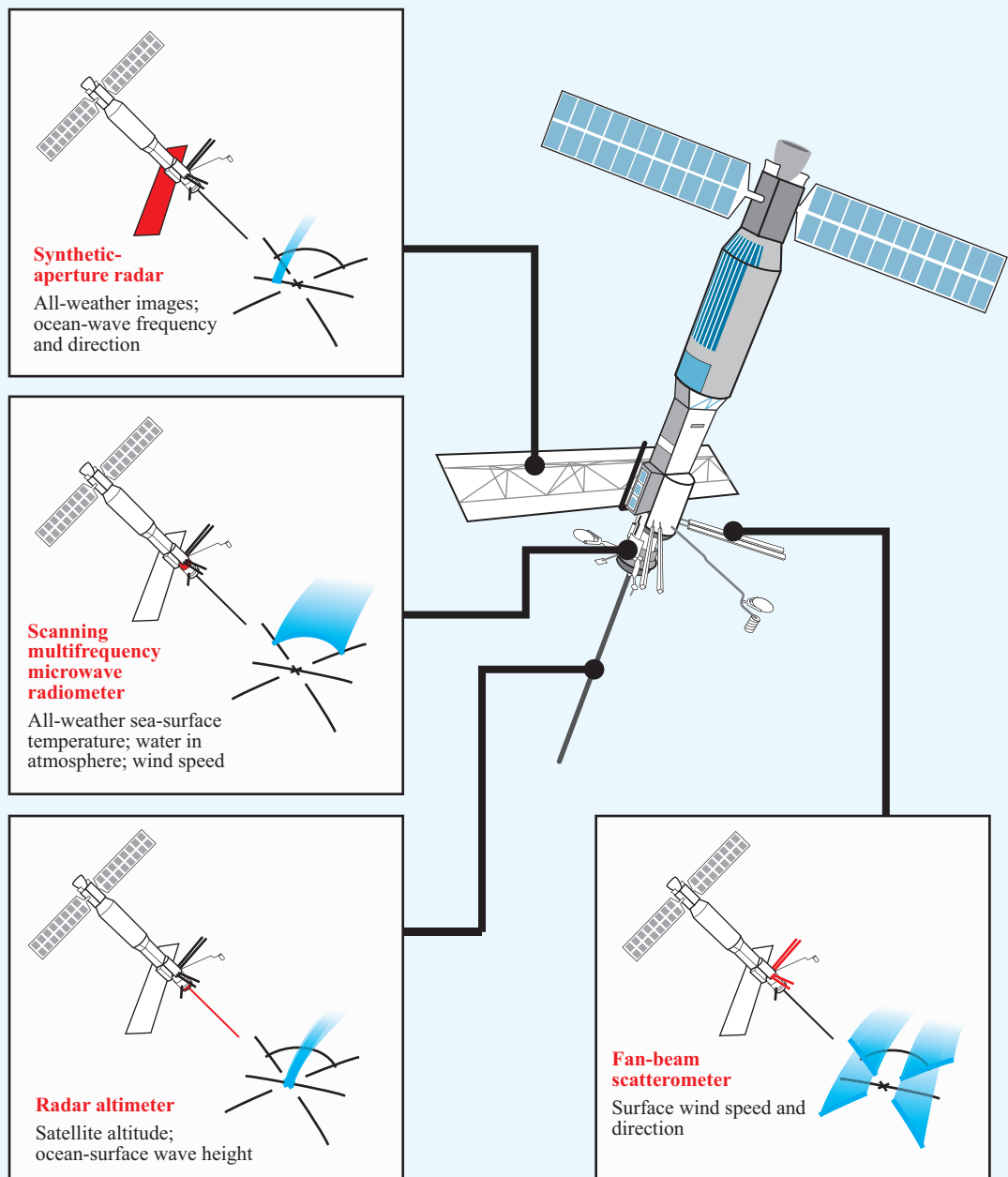
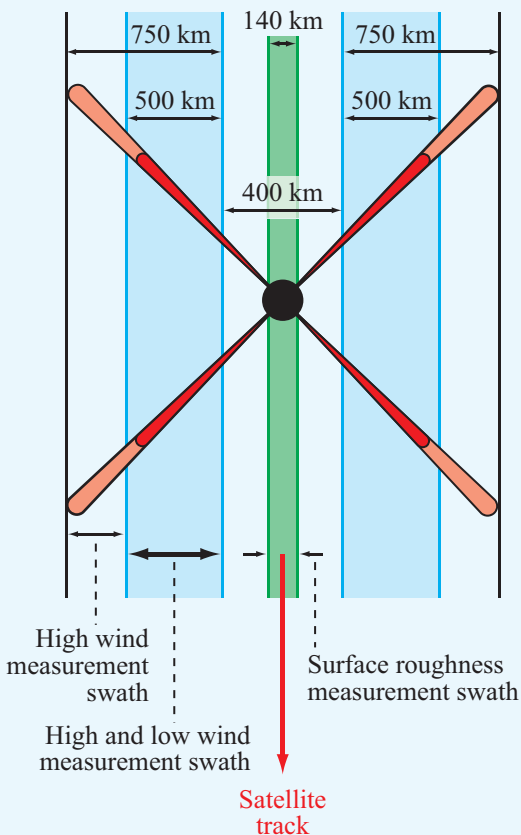
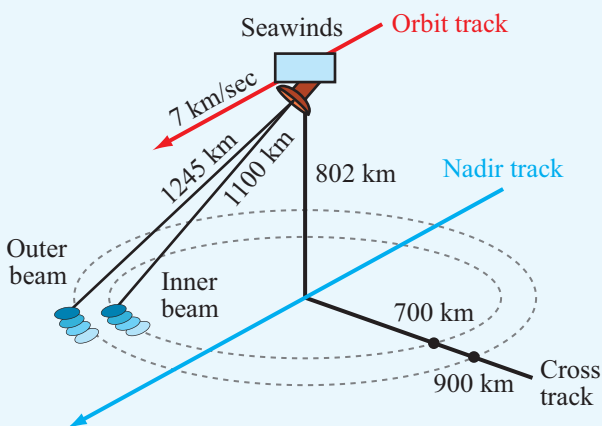


Figure 1-6: The Seasat-1 satellite, launched in 1978, with an illustration of the antenna illumination patterns for each of its microwave sensors.



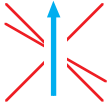
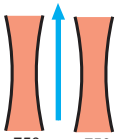
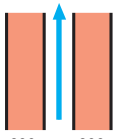
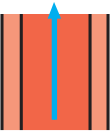
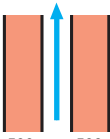
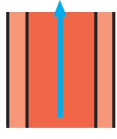
(a) Fan-beam antenna pattern (Seasat)



(b) Rotating antenna pattern (SeaWinds)

Figure 1-7: (a) Fan-beam antenna pattern for the Seasat scatterometer. (b) Rotating antenna pattern for the SeaWinds scatterometer. These are representative examples of the two major types of spaceborne scatterometers, fan-beam and scanning pencil-beam.

Table 1-2: Operational scatterometers on spacecraft.

	SASS	ESCAT	NSCAT	SeaWinds	ASCAT	OSCAT
Frequency	14.6 GHz	5.3 GHz	13.995 GHz	13.6 GHz	5.3 GHz	13.6 GHz
Antenna azimuths						
Polarizations	vv and hh	vv only	vv and hh	h-outer/h-inner	v only	v-outer/h-inner
Beam resolution	Fixed Doppler	Range gate	Variable Doppler	Pencil-beam	Range gate	Pencil-beam
Science modes	Many	SAR wind	Wind only	Wind/hi-res	Wind only	Wind/hi-res
Resolution (σ^0)	Normally 50 km	50 km	25 km	Egg: 25×35 km Slice: 6×25 km	25/50 km	Egg: 30×68 km Slice: 5×30 km
Swath, km	 ~750 ~750	 500	 600 600	 1400,1800	 500 500	 1400,1836
Incidence angles	0°–70°	18°–59°	17°–60°	46° & 54.4°	25°–65°	49° & 57°
Daily coverage	Variable	< 41%	78%	92%	65%	> 90%
Mission & dates	Seasat: 6/78–10/78	ERS-1: 92–96 ERS-2: 95–01	ADEOS-I: 8/96–6/97	QuikSCAT: 6/99–11/09 ADEOS II: 1/02–10/02	METOPD: 6/2007– METOPB: 9/2012–	OceanSat-2: 10/09–

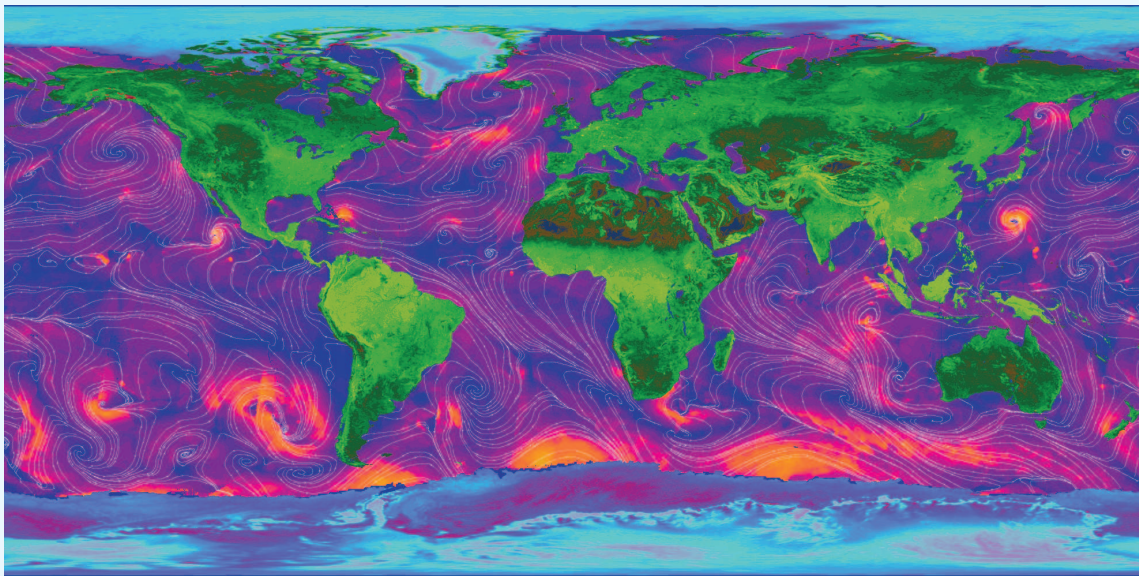


Figure 1-8: Global backscatter over land and ice derived from the SeaWinds spaceborne scatterometer. Wind speed and direction derived from ocean backscatter is shown over the ocean.

Table 1-3: History of civilian remote sensing altimeters on spacecraft.

Years	Spacecraft/Instrument	Band	Height accuracy^a (cm)
1973	Skylab/S193	Ku	–
1975	Geos3	C	–
1978	Seasat/ALT ^b	Ku	10
1983	Venera 15/16 ^b	S (8 cm)	
1985–1990	Geosat ^d	Ku	
1978–1992	Pioneer Venus-1/ORAD ^b	S (1.757 GHz)	
1990–1992	Magellan/ALT ^b	S (2.385 GHz)	3000
1991–1996	ERS-1/RA ^d	Ku	
1995–2011	ERS-2/RA ^d	Ku	
1981–2006	TOPEX-Poseidon/SSALT, NRA ^d	C, Ku	5
1997–	Cassini/ALT ^b	Ku	100
1998–2008	GFO ^d	C, Ku	
2001–	Jason-1/Poseidon-2 ^d	C, Ku	
2002–2008	Envisat/RA-2 ^d	Ku	
2006–	CloudSat/CPR ^e	W	5000
2008–	Jason-2/Poseidon-3 ^d	C, Ku	2.5
2010–	Cryosat-2/Siral ^c	Ku	
2011–	HaiYang-2 ^d	C, Ku	
2013	Saral/AltiKa ^d	Ka	

^aReported/estimated based on post processing; ^bDesigned for planetary topography measurement; ^cDesigned primarily for ice-sheet elevation and sea-ice freeboard; ^dPrimarily designed for ocean topography; ^eDesigned for cloud profiling.

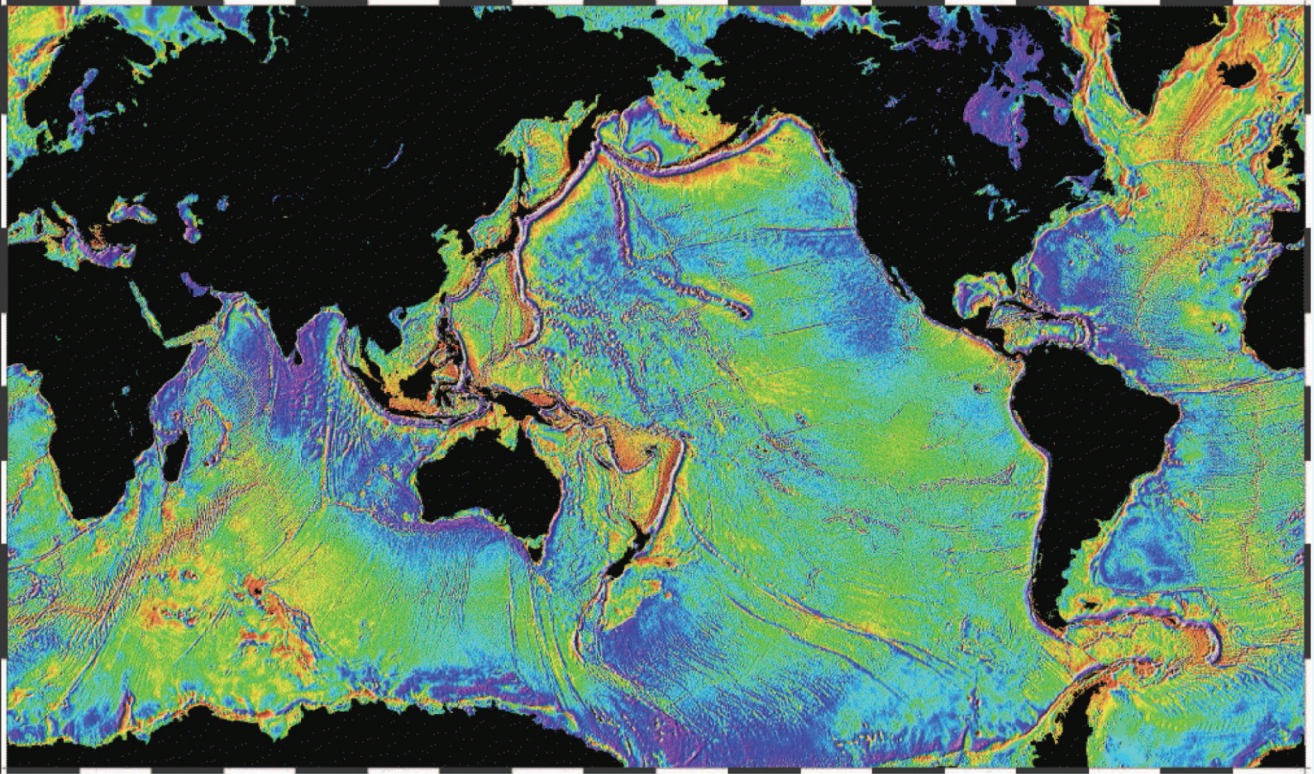
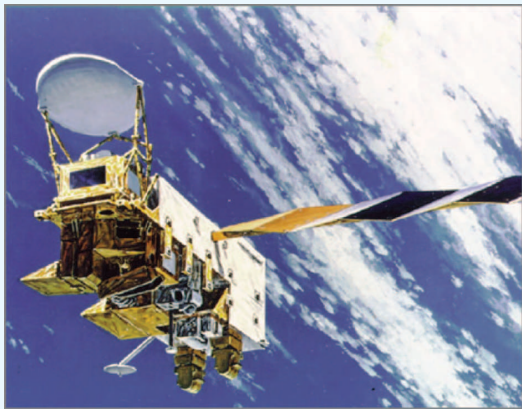


Figure 1-9: Shape of the geoid derived from spaceborne altimetry data by the ERS-1 altimeter. [Courtesy of D. Sandwell and W. Smith.]

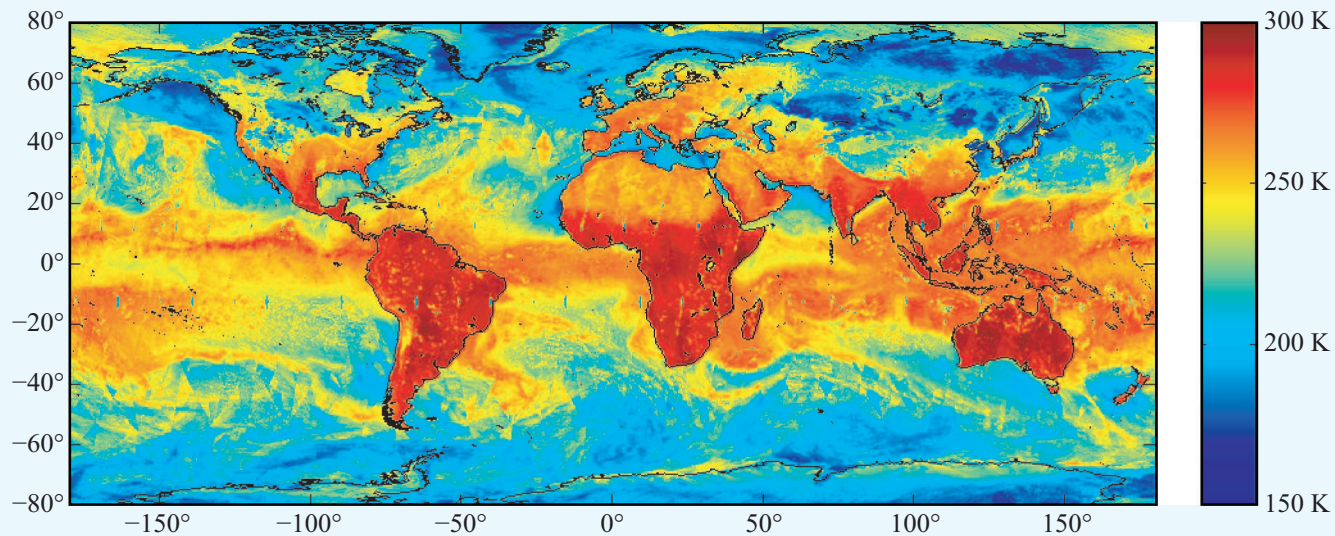
Table 1-4: History of microwave radiometry on spacecraft.

Year(s)	Spacecraft Instrument Acronym	Frequencies (GHz)	Antenna type	Swath width (km)	Resolution ^b (km)
1962	Mariner 2 Venus Flyby	15.8, 22.4	mechanically scanned parabola	planetary	1300 ^a
1968	Cosmos 243	3.5, 8.8, 22.3, 37	horns		13
1972	Nimbus 5 / ESMR	19.3	electrically scanned array	3000	25
1973	Nimbus 5 / NEMS	22.2, 31.4, 53.6, 54.9, 58.8	lens-loaded horns	nadir	200
	Skylab / S193	13.9	mechanically scanned parabola	180	16
	Skylab / S194	1.4	phased-array	nadir	115
1974	Meteor	37			
1975	Nimbus 6 / ESMR	19.3	electrically scanned array	3000	20 × 43
	Nimbus 6 / SCAMS	22.2, 31.6, 52.8, 53.8, 55.4	rotating hyperbolic mirrors	2700	150
1978	DMSP / SSM/T	50.5, 54.2, 54.3, 54.9, 58.4, 58.8, 59.4	rotating mirror	1600	175
1978, '78	TIRO N/MSU	50.3, 53.7, 55.09, 57.6	rotating mirror	2300	110
1978–87	Seasat 1 / SMMR	6.6, 10.7, 18, 21, 37	oscillating offset parabolic reflector	600	14 × 21
1978–	Nimbus 7 /SMMR	6.6, 10.7, 18, 21, 37	oscillating offset parabolic reflector	800	18 × 27
1987, '90, '91 '97, '99	DMSP /SSM/I	19.35, 22.2, 37, 85.5	spinning offset parabolic reflector	1400	15 × 13
1989	COBE	31.56, 53, 90	differential horns	full-sky	
1997–	TRMM TMI	10.7, 19.3, 21.3, 37, 85.5	spinning offset parabolic reflector	878	6
1998, '00, '02, '05	DMSP / AMSU-A/B	23.8, 89, 31.4	spinning offset reflector	1650	45
2001–11	Aqua / AMSR-E	6.9, 10.7, 18.7, 23.8, 36.5, 89	spinning offset parabolic reflector	1445	4 × 6
2001	WMAP	22, 30, 40, 60, 90	differential horns	full-sky	
2002	ADEOS-II / AMSR	6.9, 10.7, 18.7, 23.8, 36.5, 89	spinning offset parabolic reflector	1400	4 × 6
2003–	Coriolis / WindSat	6.68, 10.7, 18.7, 23.8, 37.0	spinning offset parabolic reflector	1000/400 (fore/aft)	8 × 13
2009	SMOS	1.41	synthetic aperture	1500	40
2011	Aquarius	1.413	pushbroom offset parabolic reflector	340	62 × 68
2014–15	SMAP	1.41	conically scanning mesh reflector	1000	39 × 47

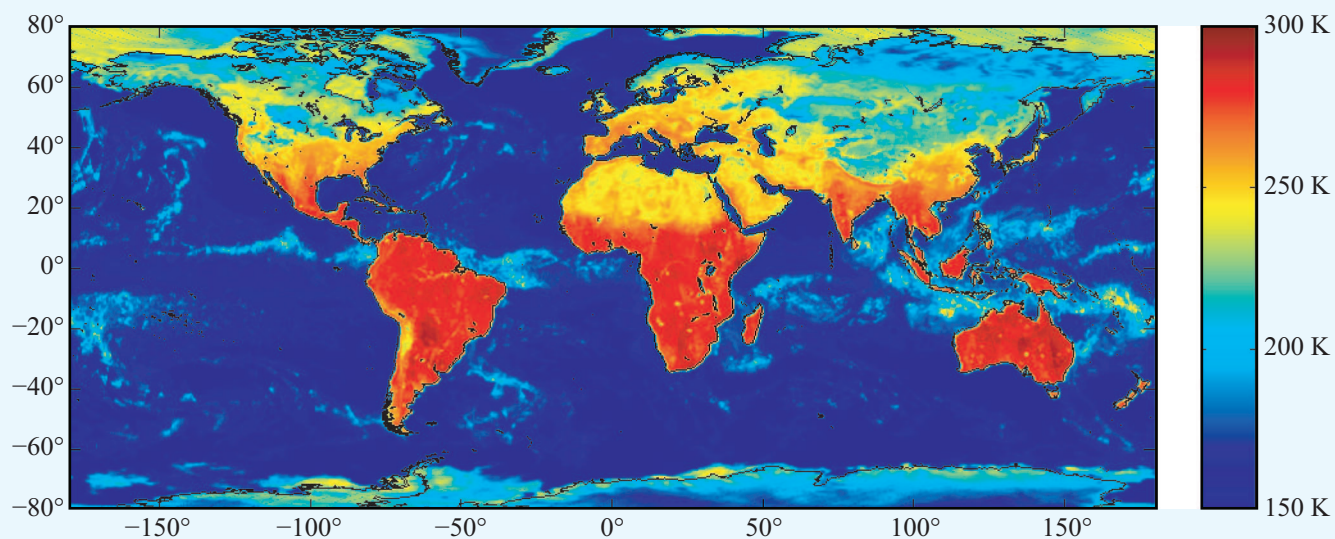
^aCourtesy of L. King, NASA/GSFC; ^bresolution of highest frequency



AMSR



(a) AMSR-E, 89.0 GHz h-polarized



(b) AMSR-E, 36.5 GHz h-polarized

Figure 1-10: Passive microwave images of the measured brightness temperature of the globe collected by the Advanced Microwave Sensor Radiometer on the Earth Observation System (AMSR-E): (a) 89 GHz h-pol, (b) 36.5 GHz h-pol.

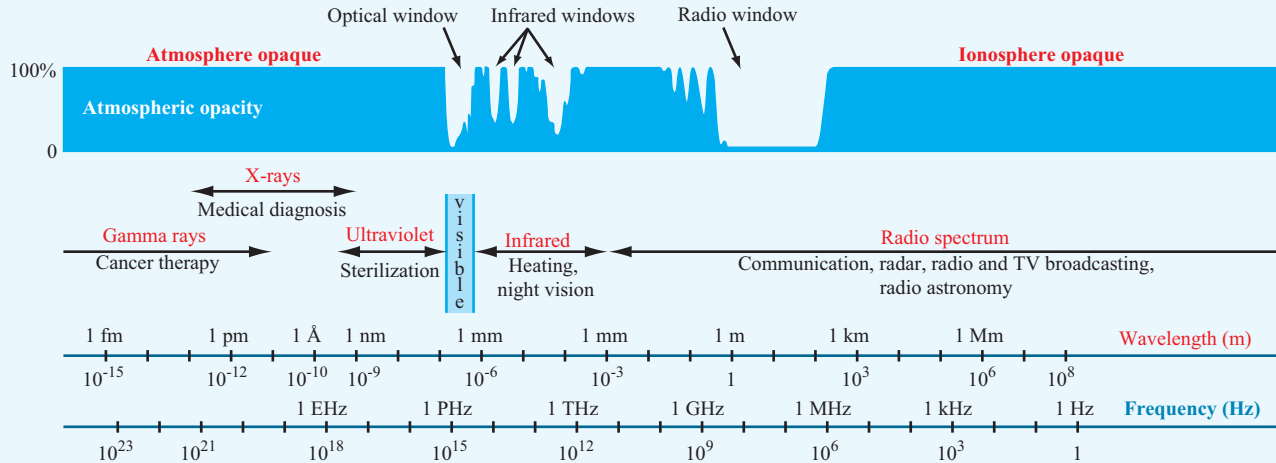


Figure 1-11: The electromagnetic spectrum. Atmospheric opacity for the Earth is shown along the top.

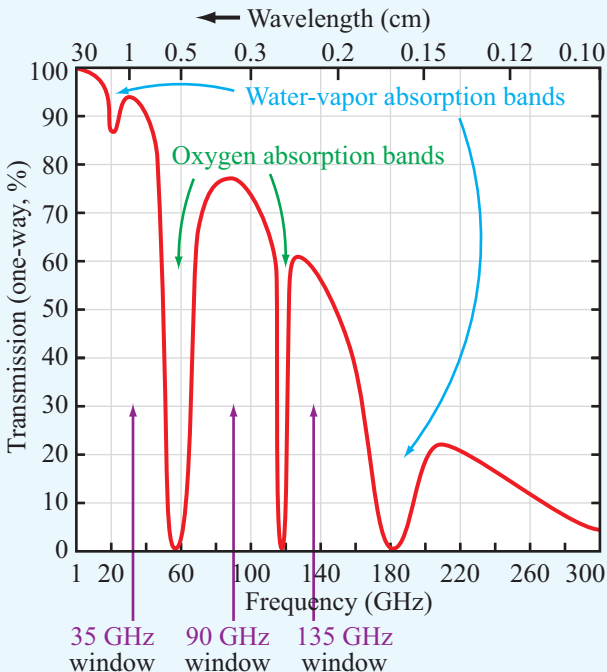


Figure 1-12: Percentage transmission through the Earth's atmosphere, along the vertical direction, under clear sky conditions.

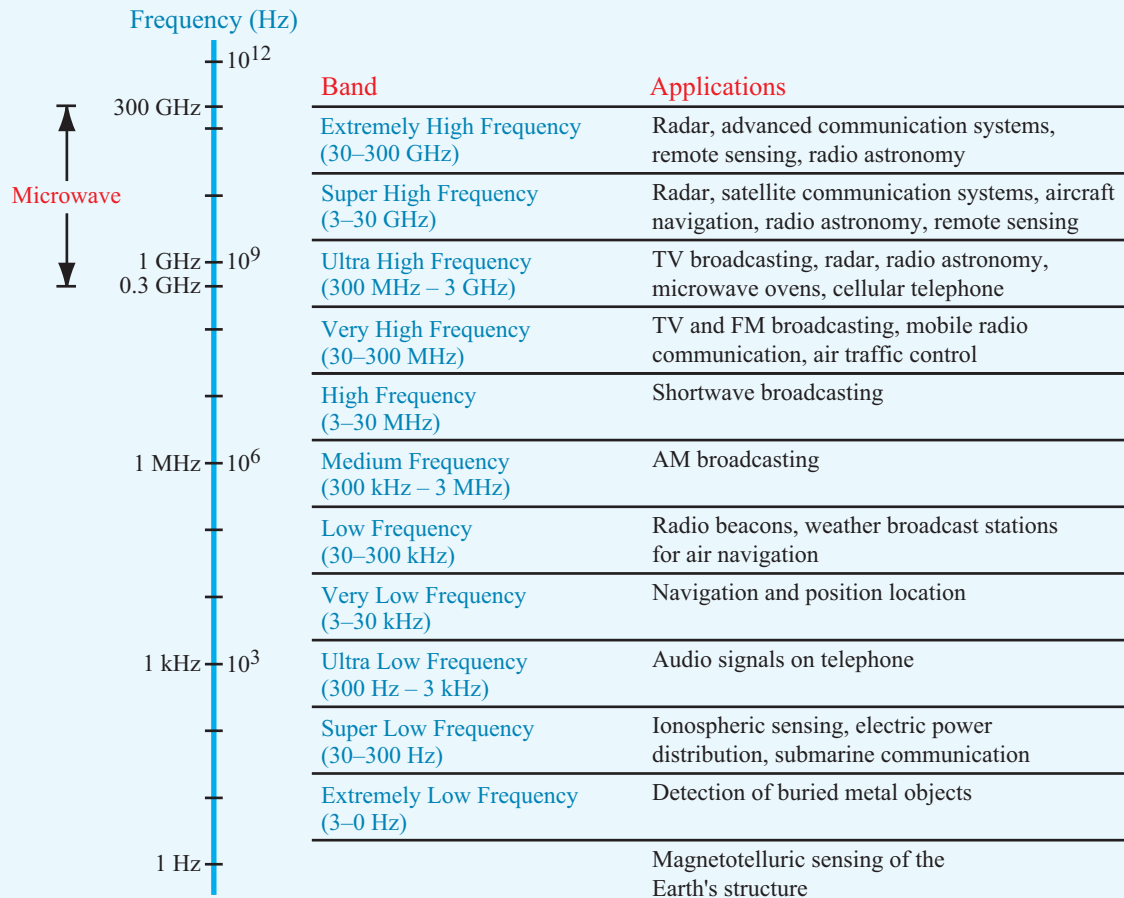


Figure 1-13: The RF spectrum and some of its applications.

Table 1-5: Common band designations (GHz).

Band	Frequency Range (GHz)
P	0.225–0.390
L	0.390–1.550
S	1.550–4.20
C	4.20–5.75
X	5.75–10.9
K	10.9–36.0
Q	36.0–46.0
V	46.0–56.0
W	56.0–100

Table 1-6: Radar frequency allocations for remote sensing (GHz) summarized from the 1992 WARC. (All are shared with other services.)

1.215–1.300, 3.1–3.3, 5.25–5.35, 8.55–8.65
9.50–9.80, 13.4–14.0, 17.2–17.3, 24.05–24.25
35.5–35.6, 78.0–79.0

Examples of Other Allocations

Radar altimeter	4.2–4.4
Doppler navigator	8.8, 13.25–13.40
Meteorological radar	5.6–5.65, 9.3–9.5
Coastal radar	5.35–5.65, 9.0–9.2, 10.0–10.55
Ship radar	5.46–5.47, 9.3–9.5, 14–14.3, 24.25–25.25, 31.8–33.4

Table 1-7: Passive-sensor frequency allocations (GHz).

0.404–0.406a	15.35–15.40a	89–92a
1.370–1.400s	21.2–21.4p	100–102p
1.400–1.427s	22.21–22.5s	105–116a
1.6605–1.6684p	23.36–24.0a	116–126p
2.600–2.640s	31.3–31.5a	150–151p
4.2–4.4s	31.5–31.8p	164–168a
4.80–4.99s	36–37p	174.5–176.5p
6.425–7.250s	50.2–50.4p	200–202p
10.60–10.68p	51.4–54.25a	217–231a
10.68–10.70a	54.25–58.2p	235–238p
15.20–15.35s	64–65a	250–242a

a: protected for radio astronomy—no transmitters allowed;
p: shared, with primary use for services having transmitters;
s: shared, with secondary use for services having transmitters.

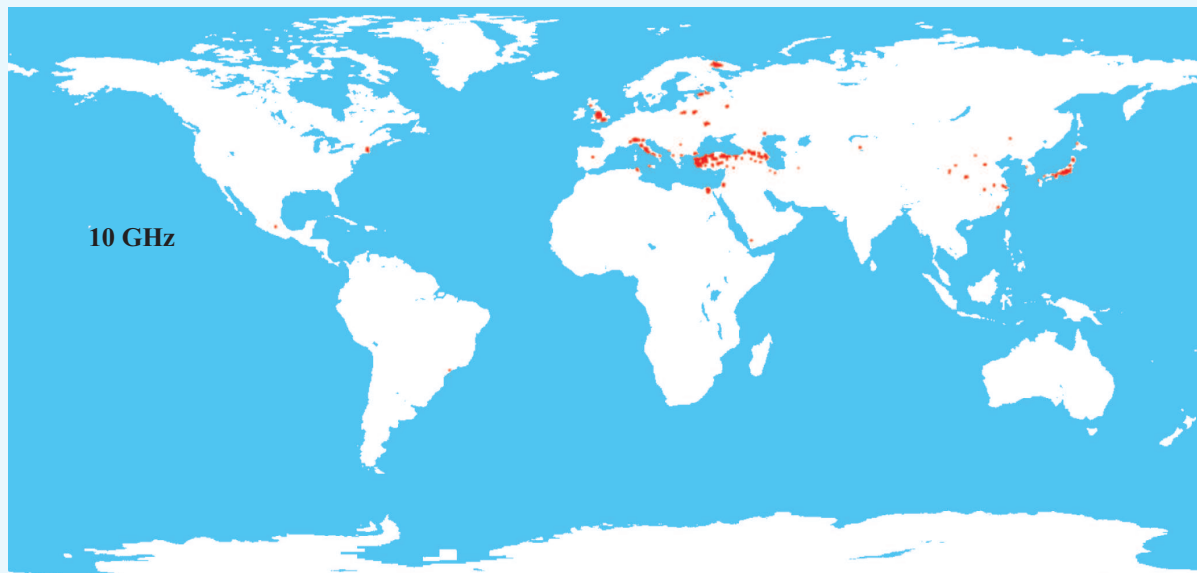


Figure 1-14: Intensity of radio frequency interference (RFI) observed by the WindSat polarimetric radiometer at 6 GHz h-pol and 10 GHz polarimetric. [Images courtesy of William Johnson and Li Li, Naval Research Laboratory.]

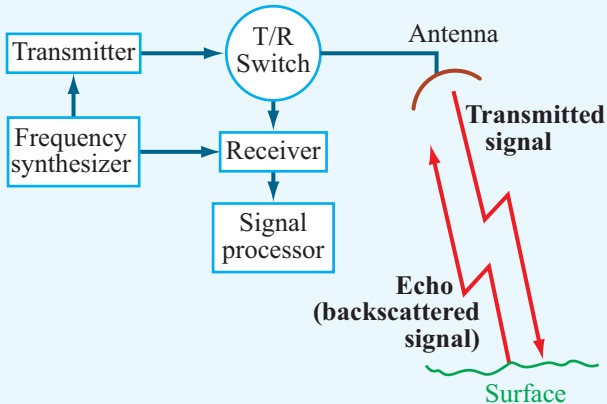


Figure 1-15: Basic elements of a remote-sensing radar system.

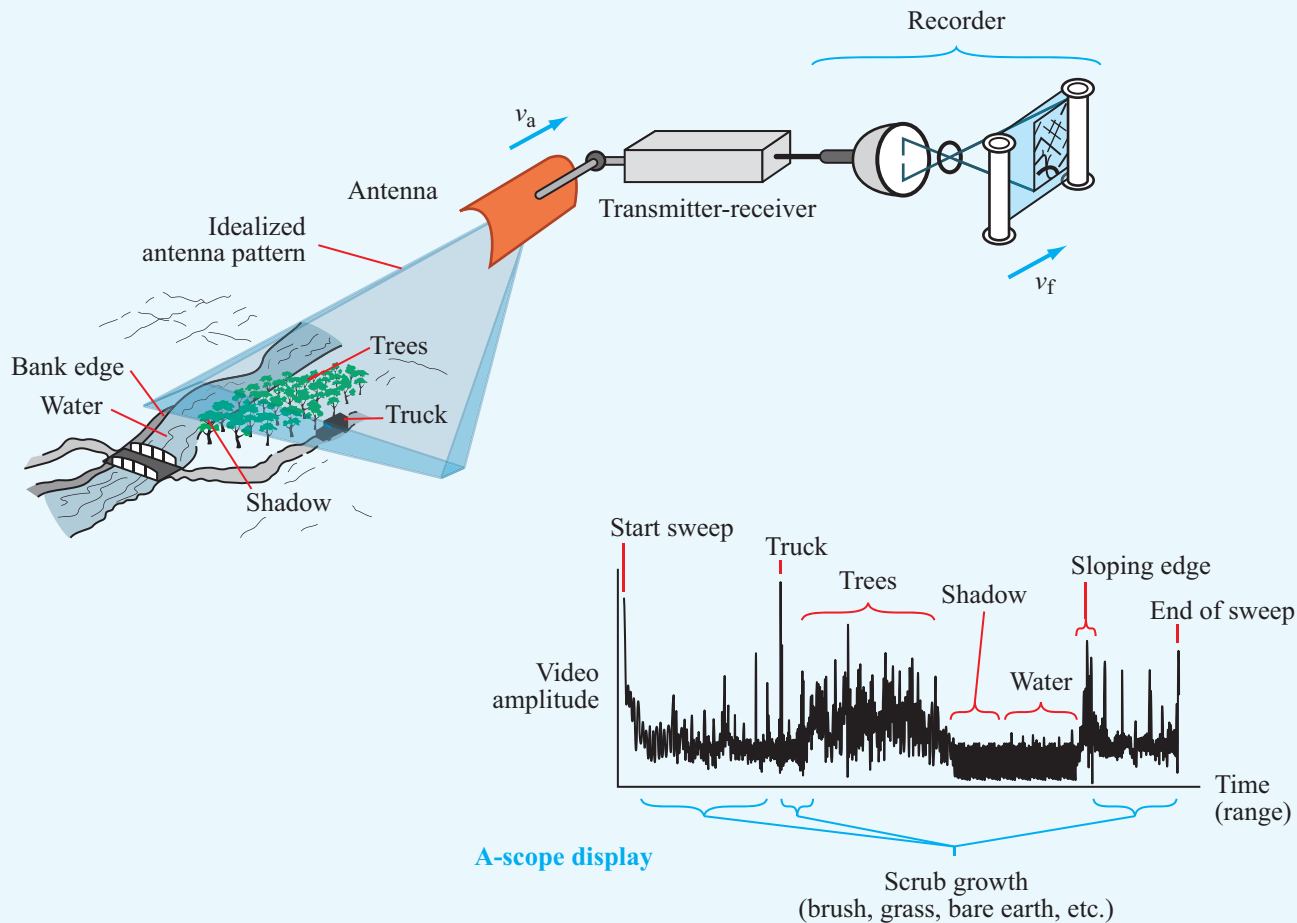


Figure 1-16: Real-aperture SLAR imaging technique.

Along-track resolution

Real-aperture $r_a = \lambda R/l$

Synthetic-aperture $r_a = l/2$

Example: 4 cm wavelength
spacecraft radar

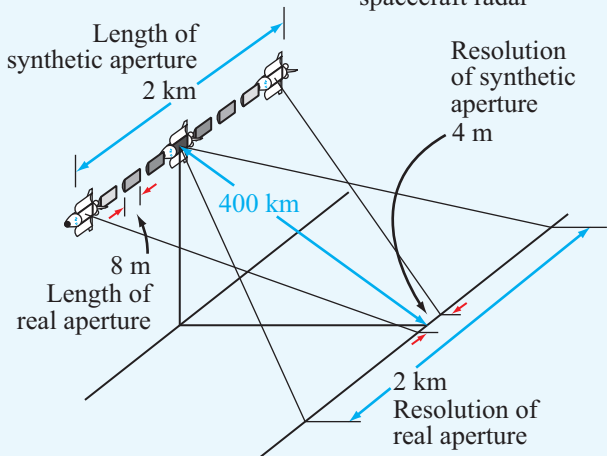


Figure 1-17: An illustration of how synthetic aperture works.

Table 1-8: Selected applications of airborne/spaceborne imaging radar in remote sensing.

Geology:

- Structure
- Lithology

Hydrology:

- Soil moisture
- Watershed mapping
- Flood mapping
- Mapping of surface water (ponds, lakes, rivers)
- Snow mapping

Agriculture:

- Crop mapping
- Agricultural-practice monitoring
- Identifying field boundaries
- Monitoring growth and harvest progress
- Identifying stress areas
- Rangeland monitoring
- Water problems—same as hydrology

Forests:

- Monitoring cutting
- Mapping fire damage
- Identifying stress areas
- Vegetation density

Cartography:

- Topographic mapping
- Land-use mapping
- Monitoring land-use changes, urban development, etc.

Polar regions (cryosphere):

- Monitoring and mapping sea ice
- Detecting and tracking icebergs
- Mapping glacial ice sheets
- Monitoring glacial changes, including measuring velocity

Ocean:

- Measuring wave spectra
- Monitoring oil spills
- Monitoring ship traffic and fishing fleets
- Wind speed and direction measurement
- Rain
- Clouds
- Measuring currents
- Undersea mapping

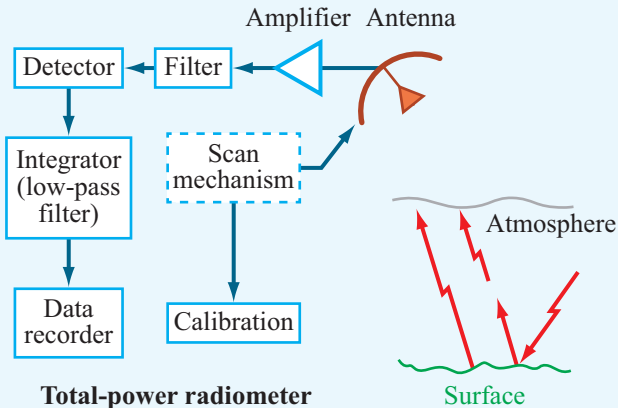


Figure 1-18: Key elements of a microwave radiometer.

Table 1-9: Selected applications of airborne/spaceborne microwave radiometers in remote sensing.

Hydrology:

- Soil moisture
- Watershed surface drainage
- Flood mapping
- Mapping of surface water (ponds, lakes, rivers)
- Snowcover extent, snow water equivalent, snow wetness

Agriculture:

- Soil moisture distribution for crop yield estimation and irrigation scheduling
- Delineation of freeze-thaw boundaries

Polar regions (cryosphere):

- Monitoring and mapping sea ice and sea-ice type
- Mapping glacial ice sheets
- Monitoring ice-sheet melt conditions

Ocean:

- Measuring surface wind speed
- Measuring surface temperature
- Measuring surface salinity
- Monitoring oil spills
- Measuring and mapping rain

Meteorology and climatology (primarily over the ocean):

- Temperature profile
- Integrated water vapor
- Water vapor profile
- Liquid water (rain)

Stratosphere, mesosphere, and lower thermosphere:

- Atmospheric temperature profile
- Magnetic field profile
- Abundance of atmospheric gases



Figure 1-19: This figure presents a Ku-band SAR image collected over Washington, D.C. The SAR illumination is from the top. Man-made features, such as buildings and bridges, stand out clearly in this radar backscatter image. The backscatter intensity of water is much smaller than that of buildings and vegetation, so water shows up black in the image. Individual trees can be distinguished in some areas, and the pattern of roads and urban development is clearly apparent. The Capitol building is left of center with the Mall extending to the right. Right of center is the Washington Monument, though only the shadow of the obelisk is readily apparent in the image. Just below the monument are the Ellipse and the White House. The Naval Research Laboratory is at the upper left. [Courtesy of Sandia National Laboratory.]

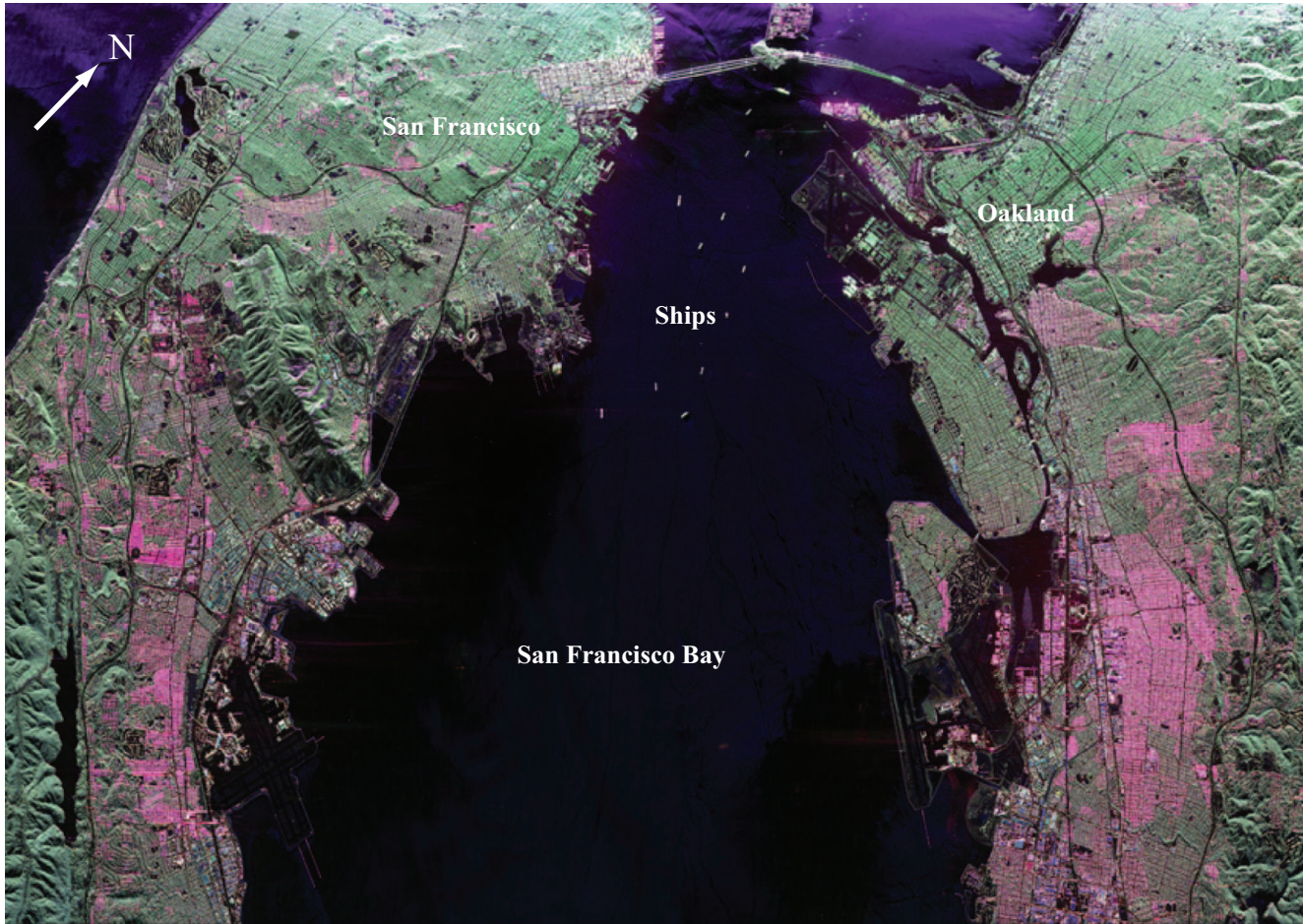


Figure 1-20: SAR image of a portion the San Francisco Bay area. Color describes the polarimetric backscatter for each pixel. The orientation of streets and roads in urban areas alters the polarimetric response in these areas, so the color is related to road orientation relative to the radar view direction, which is from the top. Water has low backscatter and appears dark. The Pacific Ocean is the dark area at the left, while San Francisco Bay is in the center. Bright objects in the bay are ships. The city of San Francisco, at the upper left, is connected to Oakland via the San Francisco-Oakland Bay Bridge, which passes through Yerba Buena Island at the top center. San Bruno Mountain is visible left of center and south of San Francisco. San Andreas Lake lies along the San Andreas fault at the western edge of the mountains at the lower left. [Courtesy of the Jet Propulsion Laboratory.]

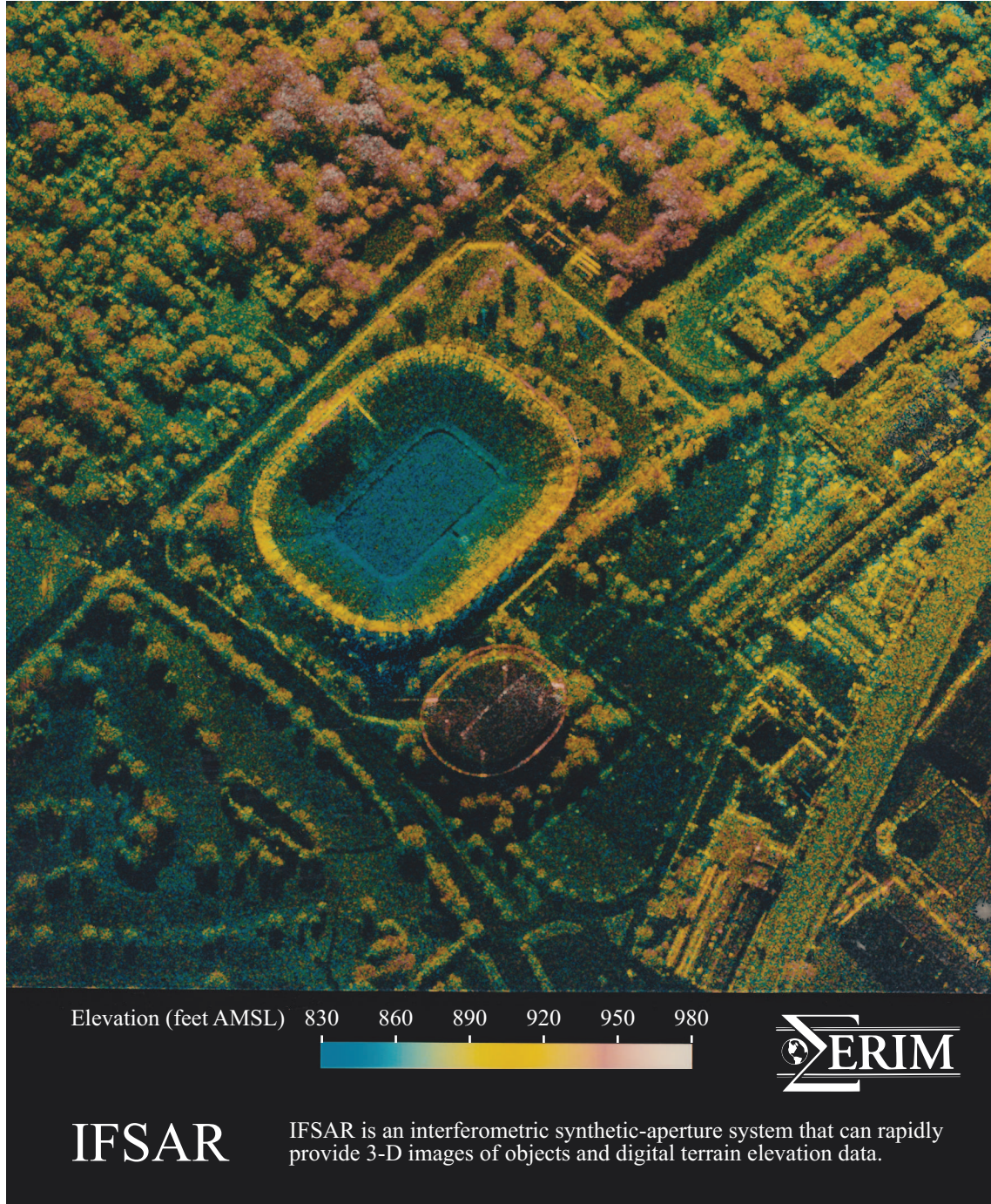


Figure 1-21: This SAR image is overlaid on a height map (color) derived using an interferometric SAR (IFSAR) technique. The image is one of the earliest demonstrations of SAR interferometry. Radar illumination is from the top. The University of Michigan football stadium is visible in the image center. Note that the playing field is depressed relative to the stands. The heights of individual trees growing on the rise in the image center can be seen. [Courtesy of the Environmental Research Institute of Michigan.]

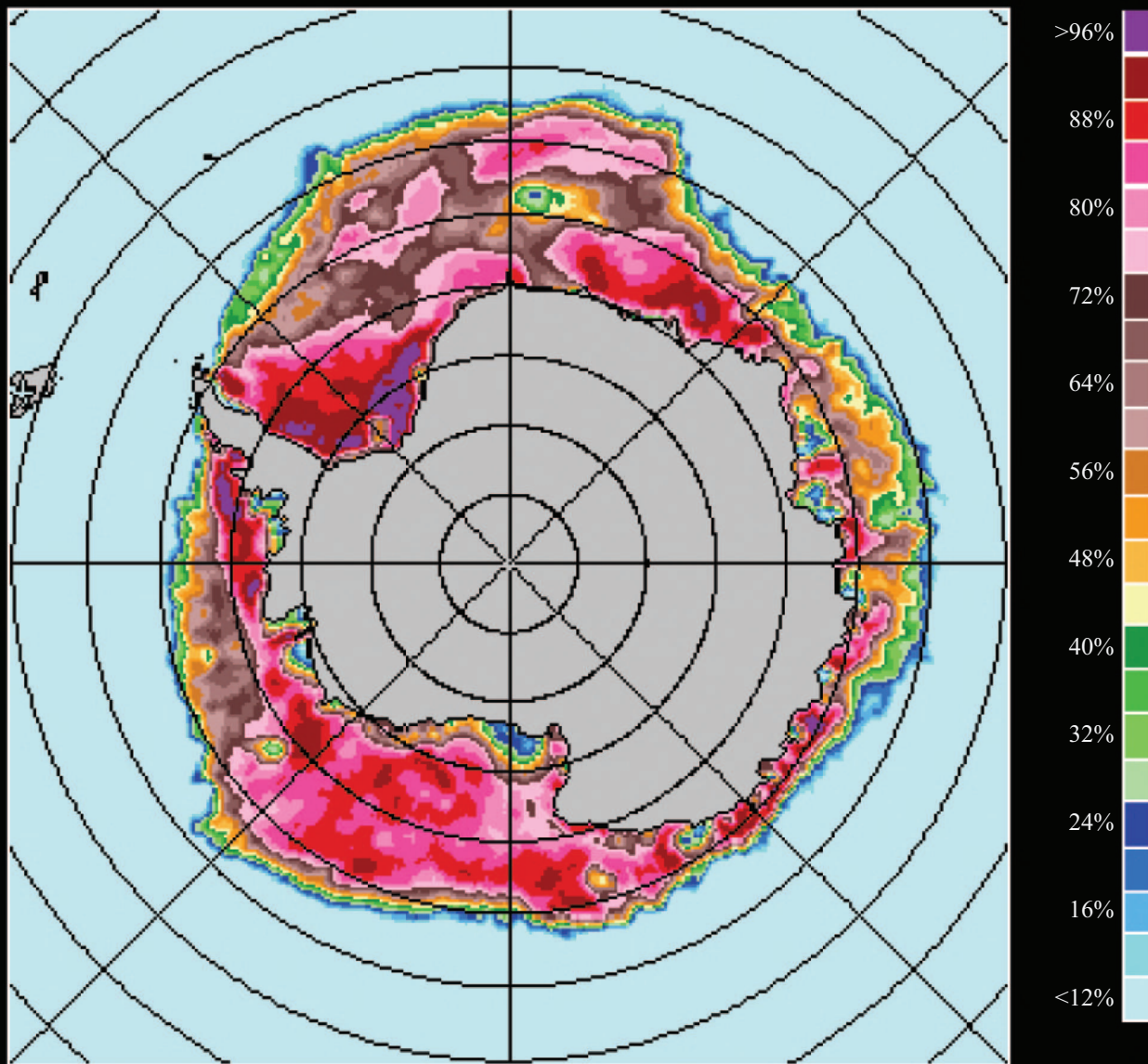


Figure 1-22: This map of sea-ice concentration in Antarctica is based on passive multichannel microwave measurements. For each pixel, the pixel color denotes the fractional amount of the pixel area covered by sea ice, with the remaining area of the pixel being open ocean. Land and near-land areas are masked off. Sea-ice coverage is an important factor in Earth's climate heat balance. [Courtesy U.S. National Sea Ice Data Center.]

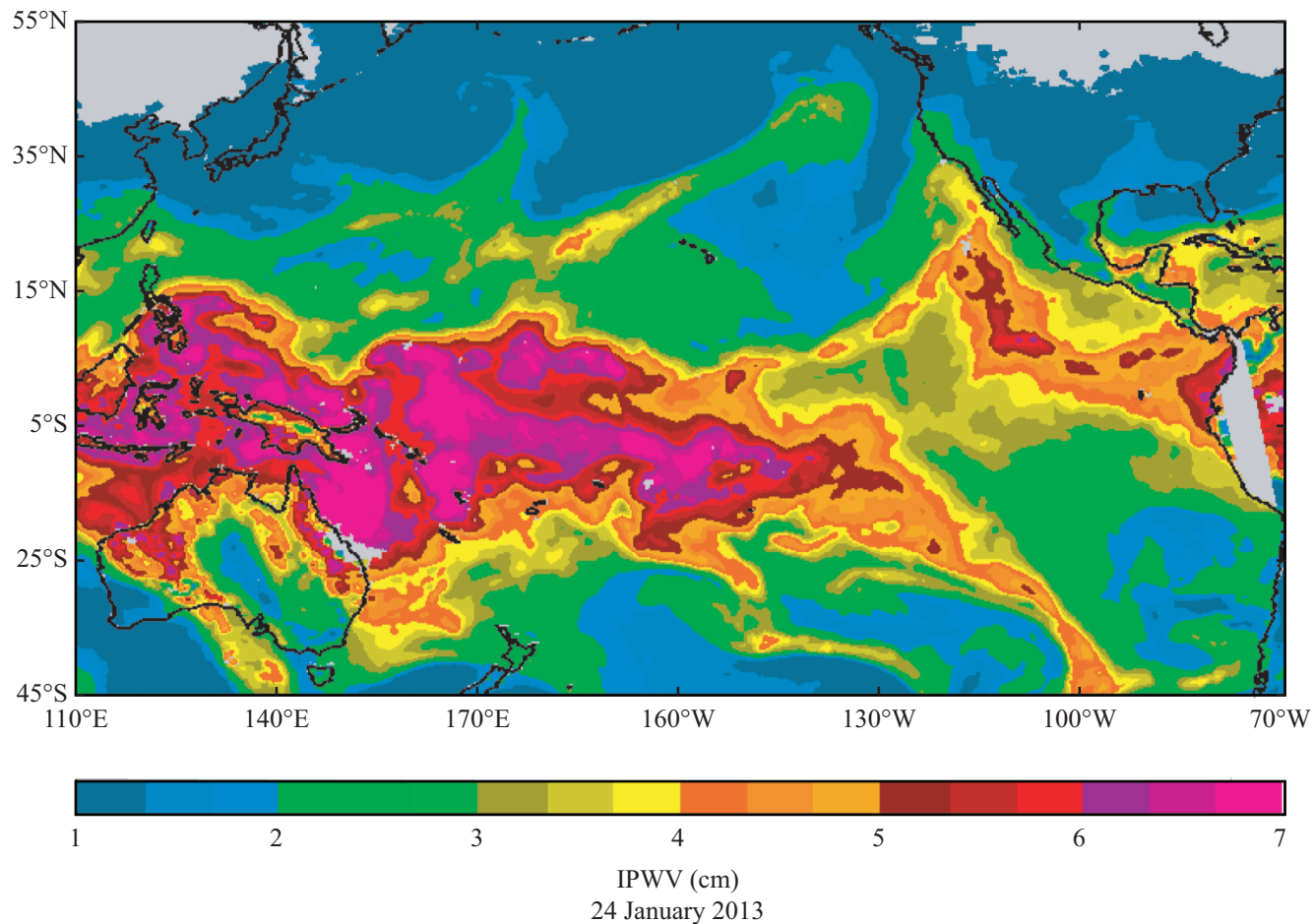


Figure 1-23: A map of integrated precipitable water vapor (IPWV). This is an important parameter used for predicting heavy precipitation. It also affects the heat balance of the atmosphere and is thus critical in climate modeling. This 6-hour-average map is derived from passive multichannel microwave observations from multiple satellite platforms and sensors whose measurements have been histogram matched (Kidder and Jones, 2007). The high values of IPWV along the southern tropical convergence zone south of the equator are indicative of high evaporation and convective rain. IPWV is measured in centimeters of equivalent precipitable water. [Courtesy of National Oceanic and Atmospheric Administration.]

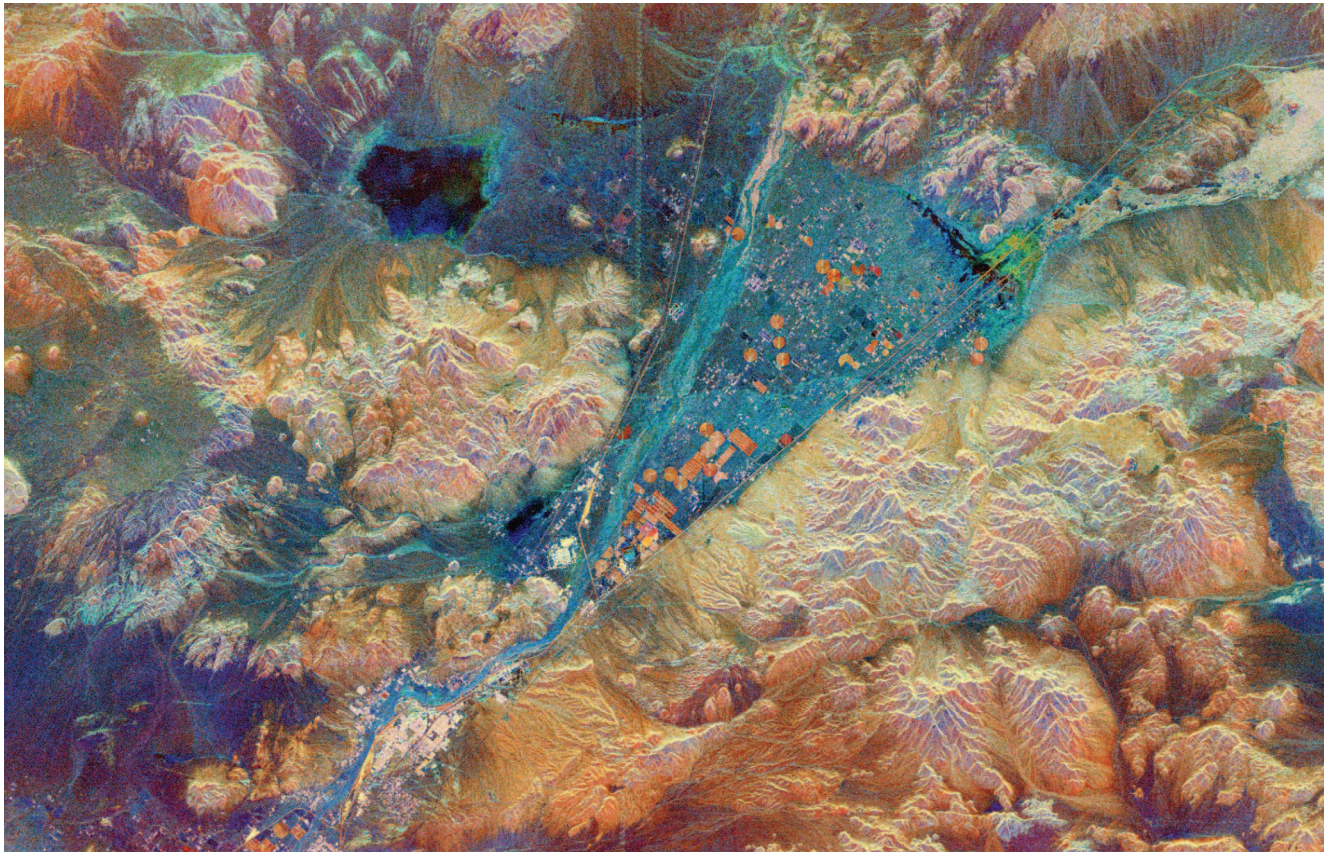


Figure 1-24: This spaceborne radar image shows part of the Mojave Desert in the vicinity of Barstow, California. [Courtesy of NASA/JPL.]